# Fuzzy System Design Code

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fis = newfis('Blood Pressure Classifier');  %Fuzzy Classifier, to determine
current state of blood pressure% 28th October ,2019

%Fuzzification

%Input 1 Coding
fis.input(1).name = 'Systolic Pressure';    % Normal Level 130/80
fis.input(1).range = [10 250];

fis.input(1).mf(1).name = 'Normal';
fis.input(1).mf(1).type = 'trimf';
fis.input(1).mf(1).params = [10 110 135];


fis.input(1).mf(2).name = 'High';
fis.input(1).mf(2).type = 'trimf';
fis.input(1).mf(2).params = [130 150 250];


% Input 2 coding

fis.input(2).name = 'Diastolic Pressure';
fis.input(2).range = [10 200];

fis.input(2).mf(1).name = 'Normal';
fis.input(2).mf(1).type = 'trimf';
fis.input(2).mf(1).params = [10 75 85];

fis.input(2).mf(2).name = 'High';
fis.input(2).mf(2).type = 'trimf';
fis.input(2).mf(2).params = [80 100 200];




%output coding
fis.output(1).name = 'Blood Pressure Classification';
fis.output(1).range = [0 100];

fis.output(1).mf(1).name = 'Normal Level';
fis.output(1).mf(1).type = 'trimf';
fis.output(1).mf(1).params = [0 20 40];

fis.output(1).mf(2).name = 'Risk Level';
fis.output(1).mf(2).type = 'trimf';
fis.output(1).mf(2).params = [35 70 100];
```

```
%Rule Evaluation

%1. IF Systolic Pressure is Normal AND  Diastolic Pressure is Normal  THEN
%Output Blood Pressure Level is Normal Level

fis.rule(1).antecedent = [ 1 1 ];
fis.rule(1).consequent = 1;
fis.rule(1).weight = 1;
fis.rule(1).connection = 1; %1 for AND; 2 for OR



%2. IF Systolic Pressure is High OR  Diastolic Pressure is High  THEN
%Output Blood Pressure Level is Risk Level

fis.rule(2).antecedent = [2 2];
fis.rule(2).consequent = 2;
fis.rule(2).weight = 1;
fis.rule(2).connection = 2;

%plotmf(fis,'input',1);

%plotmf(fis,'input',2);

%plotmf(fis,'output',1);

%Evaluation of FIS

evalfis([ 120 90],fis) % for single person assessment

round(ans)

ruleview(fis)

[Data]=xlsread('inputs_BP_Classifier','A2:B11');  %importing group data for
checking group data
input=Data;

evalfis(input,fis);


round(ans);%for rounding the ans

filename='inputs_BP_Classifier.xlsx'; %exporting group data
xlswrite(filename,ans,1,'D2')

plotfis(fis);
```